

B.Tech. Degree III Semester Supplementary Examination May 2017

ME 15-1303 MECHANICS OF SOLIDS (2015 Scheme)

Time : 3 Hours

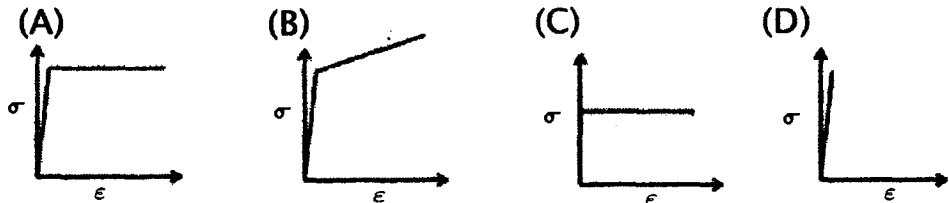
Maximum Marks : 60

PART A

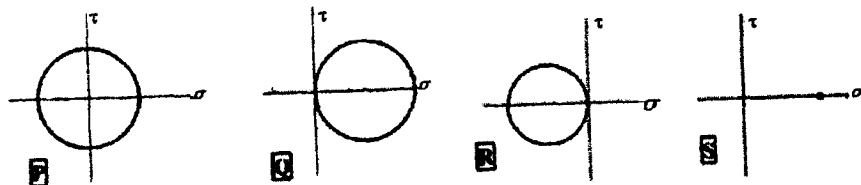
(Answer **ALL** questions)

(10 × 2 = 20)

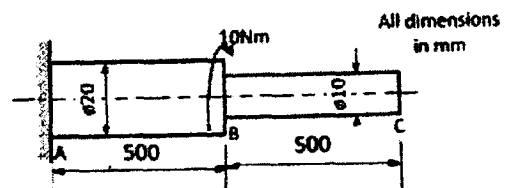
1. (a) From the following stress-strain diagrams given below (σ = stress and ϵ = strain), identify the types of materials (ductile, brittle, plastic, or elasto-plastic)



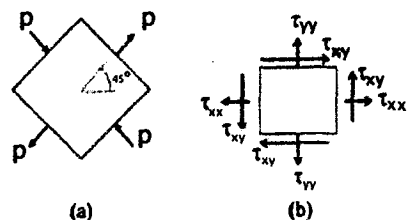
- (b) The true stress at fracture of a tensile tested specimen, having an initial diameter of 13 mm, is 700 MPa. If the diameter of specimen at fracture is 10 mm, then what is the engineering stress, in MPa, at fracture?
- (c) A circular rod of length 'L' and area of cross-section 'A' has a modulus of elasticity 'E' and coefficient of thermal expansion ' α '. One end of the rod is fixed and other end is free. If the temperature of the rod is increased by ΔT , then what are the values of stress and strain developed in the rod? Justify your answer.
- (d) Which of the following Mohr's circles of a plane-stress condition corresponds to equal bi-axial tension? Why?



- (e) Consider a stepped shaft subjected to a twisting moment applied at B as shown in the figure. Assume shear modulus, $G=77$ GPa. Find the angle of twist at C (in degrees)

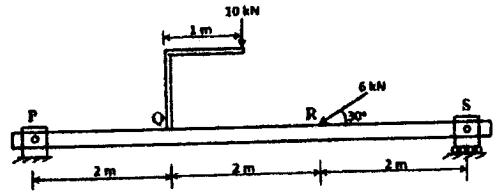


- (f) The state of stress at a point on an element is shown in figure (a). The same state of stress is shown in another coordinate system in figure (b). Determine the components (τ_{xx} , τ_{yy} , τ_{xy}).

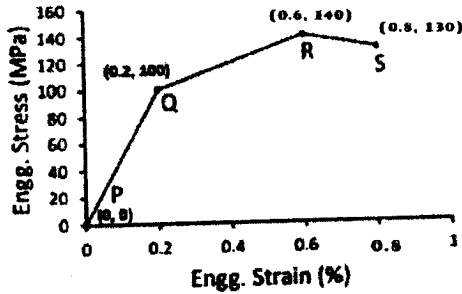


(P.T.O.)

- (g) A beam is loaded as shown in the figure. What is the bending moment, in Nm, at point R, if end P is hinged and S is roller.



- (h) A hypothetical engineering stress-strain curve shown in the figure has three straight lines PQ, QR, RS with coordinates P(0,0), Q(0.2,100), R(0.6, 140) and S(0.8, 130). 'Q' is the yield point, 'R' is the ultimate tensile stress point and 'S' the fracture point. What is the toughness of the material (in MJ/m³)?

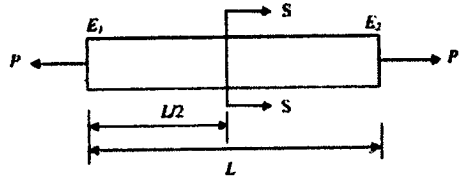


- (i) What are the assumptions used in the derivation of the flexural equation?
 (j) Explain the Tresca criterion of failure with illustration.

PART B

(4 × 10 = 40)

- II. (a) A rod of length L having uniform cross-sectional area A is subjected to a tensile force P as shown in the figure. If the Young's modulus of the material varies linearly from E₁ to E₂ along the length of the rod, find the normal stress developed at the section-SS.

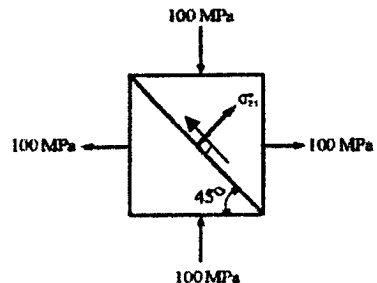


(5)

- (b) A 200 mm long, stress free rod at room temperature is held between two immovable rigid walls. The temperature of the rod is uniformly raised by 250°C. If the Young's modulus and coefficient of thermal expansion are 200 GPa and $1 \times 10^{-5} / ^\circ\text{C}$, respectively, determine the magnitude of the longitudinal stress (in MPa) developed in the rod.

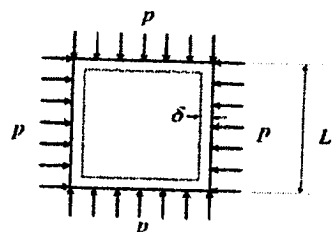
OR

- III. (a) Two triangular wedges are glued together as shown in the figure. Calculate the stress acting normal and tangential to the interface.



(5)

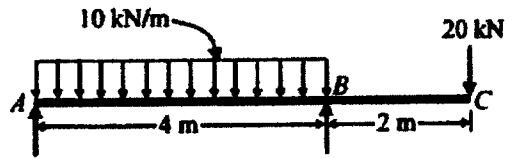
- (b) A square plate of dimension L × L is subjected to a uniform pressure load p = 250 MPa on its edges as shown in the figure. Assume plane stress conditions. The Young's modulus E = 200 GPa. The deformed shape is a square of dimension L - 2δ. If L = 2 m and δ = 0.001 m, find the Poisson's ratio of the plate material.



(5)

(Contd....3)

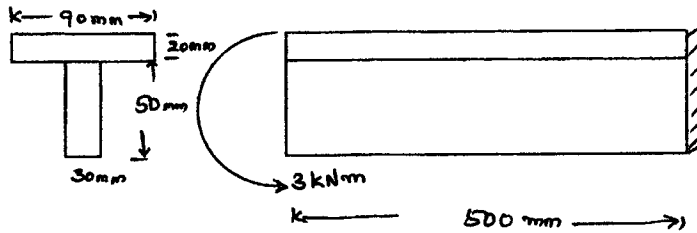
- IV. For the overhanging beam shown in figure, draw the shear force and bending moment diagram. (10)



OR

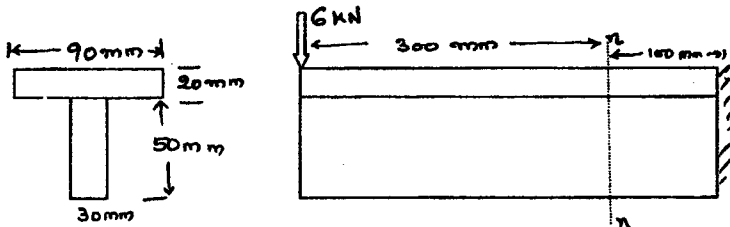
- V. A hollow shaft of 1 m length is designed to transmit a power of 30 kW at 700 rpm. The maximum permissible angle of twist in the shaft is 1° . The inner diameter of the shaft is 0.7 times the outer diameter. The modulus of rigidity is 80 GPa. Find the outside diameter (in mm) of the shaft. (10)

- VI. A cast iron machine part is acted upon by 3 kN.m couple as shown in the figure. Knowing that $E = 175$ GPa and neglecting the effect of fillets, determine (i) the maximum tensile and compressive stresses in the casting, (ii) the radius of curvature of the casting (10)

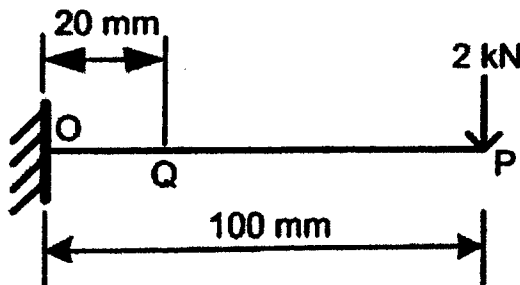


OR

- VII. A machine part has T-shaped cross section and is acted upon in its plane of symmetry by the single force as shown in figure. Determine (i) the maximum compressive stress at section n-n, (ii) the maximum shearing stress. (10)



- VIII. A cantilever beam with square cross-section of 6 mm side is subjected to a load of 2 kN normal to the top surface as shown in the figure. The Young's modulus of elasticity of the material of the beam is 210 GPa. Determine the magnitude of slope (in radian) at Q (20 mm from the fixed end) and the deflection at P. (10)



OR

- IX. Derive the Euler equation for critical load for a column with both ends hinged and carrying a central load. (10)